

Exam Project Presentation

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Exam Structure and Dates

The exam is composed by two parts:

- **Written Test (50%):**
 - 10 questions with multiple answers
 - 2 open questions
 - 1 problem
- **Project Presentation (50%):**
 - Individual or Group

Dates:

- Written Test: **8/01/2025, 9/02/2025**
- Project Presentation: **14/01/2025, 11/02/2025**
 - Remember to send the project code and presentation slides 2 days before the exam date on a shared folder.

Project Presentation

- A report (presentation or document) must be presented at the exam, together with the implemented code (provided separately, not in the report);
- During the exams (project presentation) the students will be asked to show the results by means of simulations or videos;
- A clear organization of the project and the related code is highly recommended;
- Available ROS packages can be exploited for the project implementation.

Remember to send the project code and presentation slides 2 days before the exam date on a shared folder.

Two ROS2 packages have been prepared for the exam project:

- **tiago_exam**: contains the launch file for launching the simulation environment for the exam project;
- **tiago_exam_worlds**: contains the gazebo environment for the exam project.

Both packages should be placed in tiago_ws/src folder.

Project - Group Number

The final project concerns:

- Robot navigation exploiting the knowledge acquired in Module 1;
- Robot manipulation exploiting the knowledge acquired in Module 2;

As robot platform for both parts, you are going to use the Tiago robot.

Each group will have a specific simulation environment assigned.

First, you need to form the groups by exploiting the available section on the virtuale course platform.

 Final Exam Project Groups

[Scelta gruppo](#) [Impostazioni](#) [Visualizza 0 risposte](#) [Altro ▾](#)

Final Exam Project Groups

Spunta come completato

Visualizza 0 risposte

Scelta	Gruppo	Visualizza descrizione	Membri / Capienza	Membri del gruppo
				Visualizza membri del gruppo
☐	Group 20		0 / 3	
☐	Group 22		0 / 3	
☐	Group 23		0 / 3	
☐	Group 24		0 / 3	

Project - Running

Given your group number N (the one from virtuale), you need to send an email specifying your group number to alessio.caporali@unibo.it to receive the specific configuration files for your simulation environment.

For debugging purposes, you can launch the **example environment** by executing:

```
ros2 launch tiago_exam tiago_exam.launch.py world_name:=example
```

Remember to replace example with the world name assigned to your group (you will receive it by email).

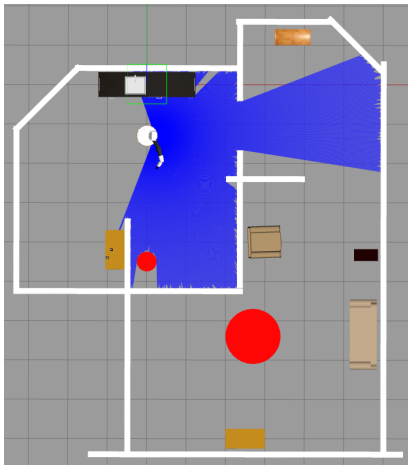
For activating the manipulation stack (Moveit), you can append the following line to the launch command:

```
moveit:=true
```

Task 1: Map Generation

The students are asked to make the robot able to move autonomously in the environment to create a map covering the whole scenario.

Existing packages can be used. Example of the simulation environment:



Task 1: Map Generation

For launching the navigation stack with SLAM (for map generation), you need to execute:

```
ros2 launch tiago_2dnav tiago_nav_bringup.launch.py  
is_public_sim:=false rviz:=True slam:=True
```

For saving the map generated by the SLAM algorithm:

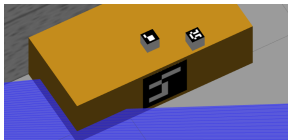
```
ros2 run nav2_map_server map_saver_cli -f map_name
```


Task 2: Navigation

After the mapping of the scenario (Task 1), the students are asked to make the robot able to navigate in the environment reaching the pick and place locations of the manipulation task (next task).

The alignment between the map and the navigation system must be performed with the robot initialized in a random position (different from the one used for map generation).

The pick and place locations of the manipulation task are represented in the figure below by the dark yellow objects. The robot must be able to discover and reach those locations by exploiting the aruco markers placed on their side. **An autonomous discovery and navigation to those locations should be implemented.**



(a) Pick location: aruco marker with ID 26



(b) Place location: aruco marker with ID 238

Task 2: Navigation

First, you need to launch the navigation stack with the map generated in Task 1. You can do this performing these steps:

- create a new folder somewhere in your environment, e.g. *my_map*. The folder should contain the generated *map.pgm* and *map.yaml* files.
- launch the navigation with the generated map:

```
ros2 launch tiago_2dnnav tiago_nav_bringup.launch.py  
is_public_sim:=false rviz:=True map_path:=path_my_map
```

Tip: the alignment between the map and the navigation system can be checked by looking at the *amcl_pose* topic.

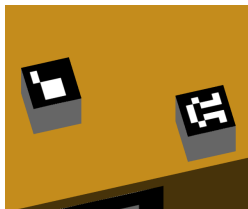
Aruco markers side length 25 cm.

Task 3: Pick and Place

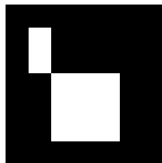
After the navigation task (Task 2), the students are asked to make the robot able to transport each cube from the pick location to the place location, following the specified sequence:

- 1 transport of the aruco marker with ID 63;
- 2 transport of the aruco marker with ID 582;

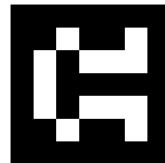
Cube dimension 7 cm, aruco marker side length 7 cm.



(a) Cubes



(b) ID 63



(c) ID 582

General Recommendations

In addition to the standard nodes/launch files to execute, **we expect to have 3 launch files to launch each of the 3 tasks.**

Some general recommendations for the project implementation:

- The use of non-standard/questionable procedures will be negatively evaluated, e.g. calling ROS topic or services as subprocesses within another node. Please use ROS functionalities as they are intended to be used (publishers/subscribers, service clients/servers..);
- The robot clearly missing the cube during grasping or dropping it outside the place location will be negatively evaluated;
 - The gripper closing/opening procedure must be clearly visible during the simulation;
 - It is possible to use the *Link Attacher* plugin to facilitate the grasping procedure BUT reasonable alignment between the end-effector and the cube must be performed;
 - Trying to avoid the use of the plugin will be positively evaluated (e.g. modifying the simulation friction, etc..);
- The use of hardcoded positions for pick and place will be negatively evaluated; The robot must be able to autonomously discover the pick and place locations by exploiting the aruco markers;

After a CTRL+C on a ROS2 launch command, the gazebo simulation may fail to exit properly. **This may cause unexpected behaviors if we try to launch the simulation again.**

To verify if the simulation is still running, we can use the following command:

```
ps -ef | grep ros
```

If the simulation is still appended to the process list, you will see an output similar to the following:

```
student 12433 11844 0 14:05 pts/2 00:00:01 /usr/bin/python3  
/opt/ros/humble/bin/ros2 launch tiago_gazebo tiago_gazebo.launch.py
```

To kill the process, you can use the following command:

```
sudo kill -9 12433
```